

## Project Initialization and Planning Phase

|               |                                                             |
|---------------|-------------------------------------------------------------|
| Date          | 29 June 2026                                                |
| Team ID       |                                                             |
| Project Name  | OptiCrop: Smart Agricultural Production Optimization Engine |
| Maximum Marks | 3 Marks                                                     |

### Define Problem Statements (Customer Problem Statement Template):

Create a problem statement to understand your customer's point of view. The Customer Problem Statement template helps you focus on what matters to create experiences people will love. A well-articulated customer problem statement allows you and your team to find the ideal solution for your customers' challenges. Throughout the process, you'll also be able to empathize with your customers, which helps you better understand how they perceive your product or service.

#### Example:

|                                         |                                           |                                                             |                                                                                            |                                               |
|-----------------------------------------|-------------------------------------------|-------------------------------------------------------------|--------------------------------------------------------------------------------------------|-----------------------------------------------|
| <i>I am</i>                             | <i>I'm trying to</i>                      | <i>But</i>                                                  | <i>Because</i>                                                                             | <i>Which makes me feel</i>                    |
| Farmer seeking higher crop productivity | Choose the most suitable crop for my farm | I don't know which crop will give the best yield and profit | Soil nutrients and weather conditions change every season, making crop selection difficult | Worried about crop failure and financial loss |

| Problem Statement (PS) | I am (Customer)                     | I'm trying to                                         | But                                                               | Because                                                                                                                                             | Which makes me feel                                                                                  |
|------------------------|-------------------------------------|-------------------------------------------------------|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| PS - 1                 | A farmer cultivating seasonal crops | Select the most suitable crop for the upcoming season | I am not sure which crop will grow well and provide better profit | Soil nutrients, rainfall, temperature, humidity, and pH vary across seasons, and I do not have a reliable way to analyze all these factors together | Confused about making the right farming decision and worried about low crop yield and financial loss |